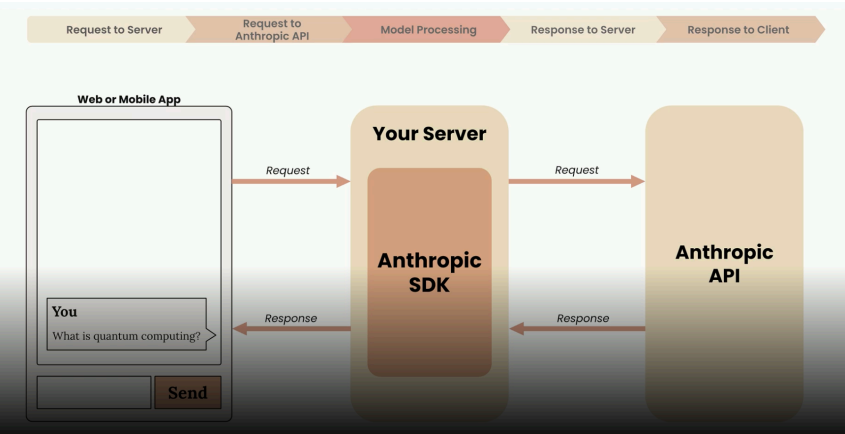


# Overview

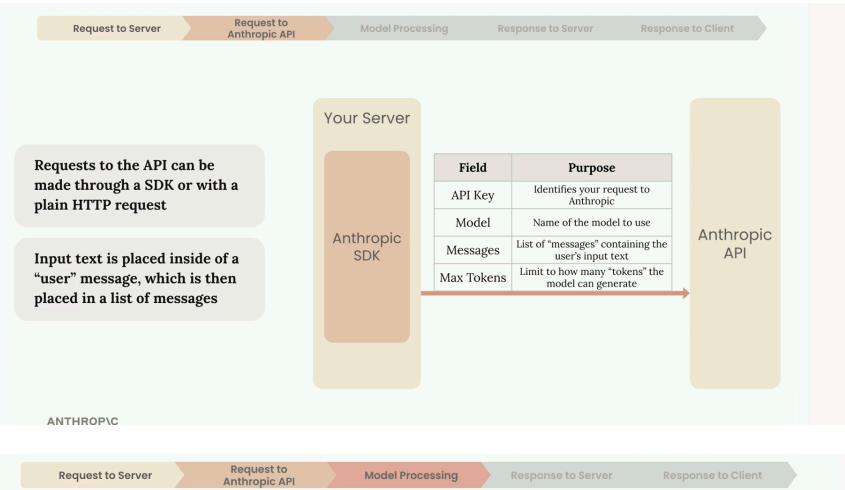
Thursday, 25 June 2026 11:20 AM

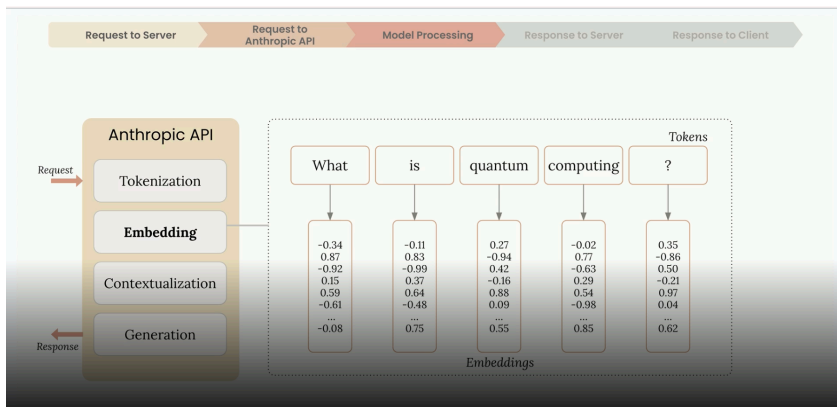
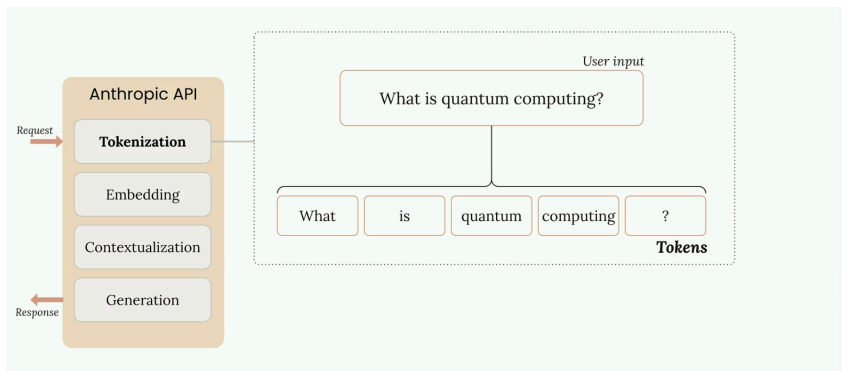
| Feature / Metric    | Claude Opus                                                                                                                                                                                                                                                                        | Claude Sonnet                                                                                                                                                                                                                                                    | Claude Haiku                                                                                                                                                                                                                                                                     |
|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Description         | Highest level of intelligence and advanced capabilities.                                                                                                                                                                                                                           | Intelligent model balancing quality, speed, and cost.                                                                                                                                                                                                            | Most cost-efficient and latency-optimized model.                                                                                                                                                                                                                                 |
| Cost                | High                                                                                                                                                                                                                                                                               | Medium                                                                                                                                                                                                                                                           | Low                                                                                                                                                                                                                                                                              |
| Comparative Latency | Moderate                                                                                                                                                                                                                                                                           | Fast                                                                                                                                                                                                                                                             | Fastest                                                                                                                                                                                                                                                                          |
| Supports Reasoning  | Yes                                                                                                                                                                                                                                                                                | Yes                                                                                                                                                                                                                                                              | No                                                                                                                                                                                                                                                                               |
| Best Used For       | <ul style="list-style-type: none"><li>• Advanced software development &amp; large-scale architecting</li><li>• Complex, long-running tasks requiring strategic planning &amp; multi-step problem-solving</li><li>• Research, deep analysis, and advanced reasoning tasks</li></ul> | <ul style="list-style-type: none"><li>• Common coding &amp; application development tasks</li><li>• Document creation, editing, and content marketing</li><li>• Data analysis, visualization projects, and image analysis</li><li>• Process automation</li></ul> | <ul style="list-style-type: none"><li>• Quick code completions and real-time suggestions</li><li>• High-volume content moderation and filtering</li><li>• Fast data extraction and categorization</li><li>• Language translation &amp; rapid Q&amp;A retrieval systems</li></ul> |

## Accessing the API :0-

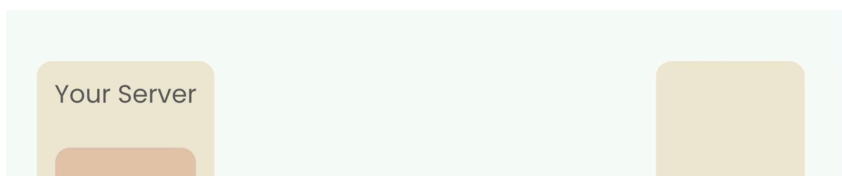
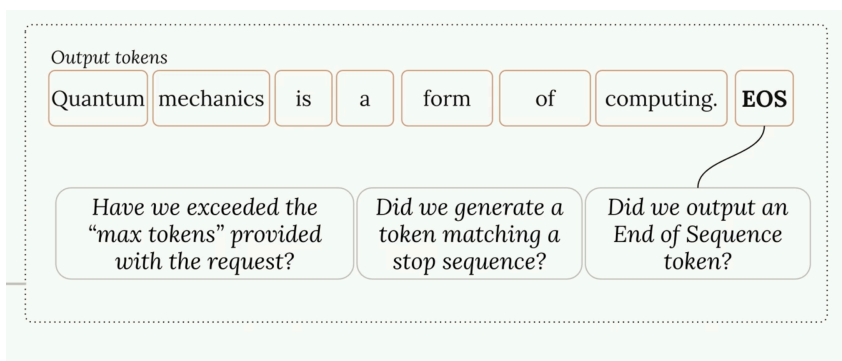
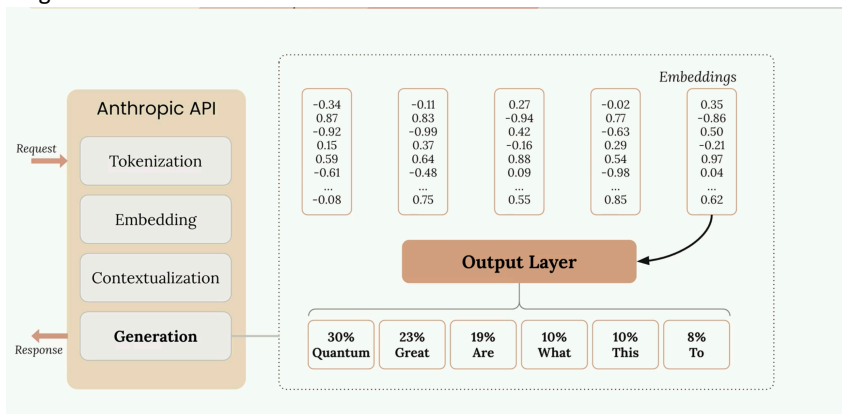


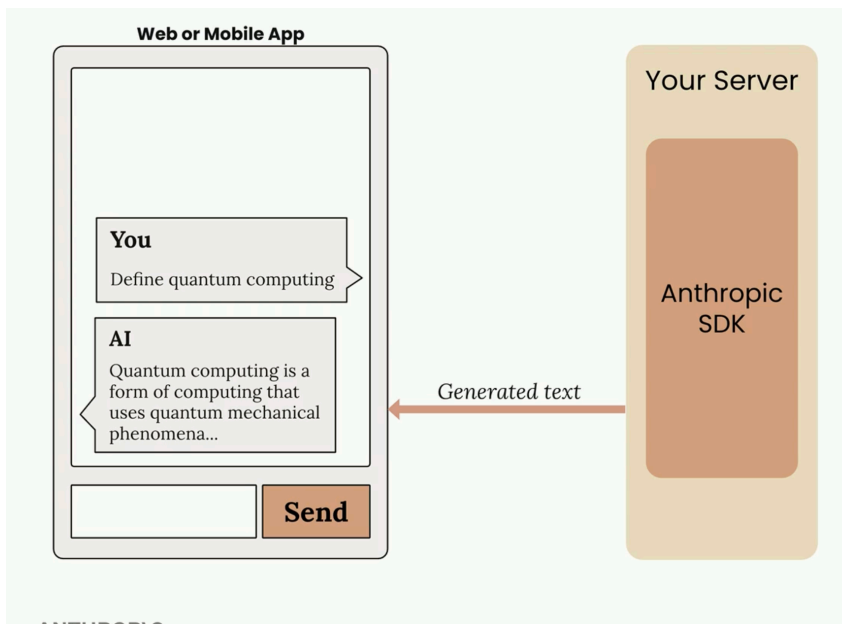
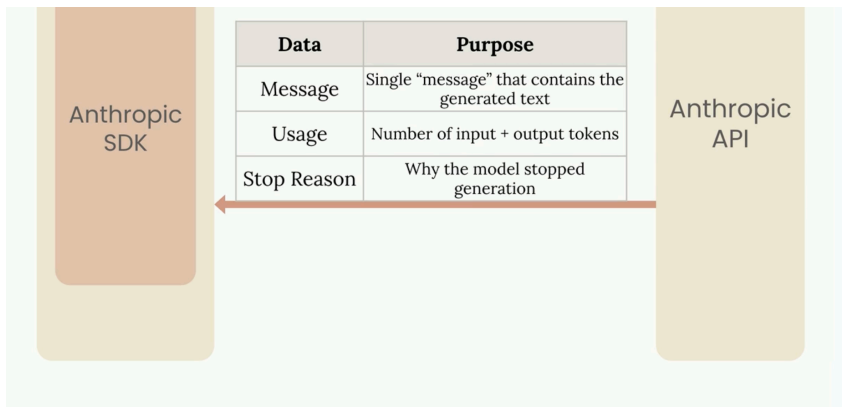
- Requests to Anthropic should not be made directly from client code
- API requests requires a secret API key, which shouldn't be exposed to the client





In Contextualization each embedding is adjusted based on all embeddings around it which makes more sense given its neighbours

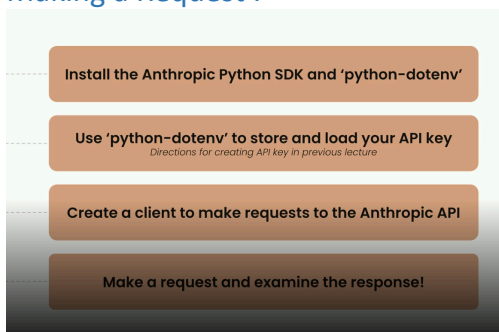




## Generating an API key :-

- Step One: Navigate to the Anthropic API Console
- Step Two: Click the 'Get API Keys' button
- Step Three: Click the 'Create Key' button
- Step Four: Enter a workspace and name for your key
- Step Five: Copy the Key

## Making a Request :-

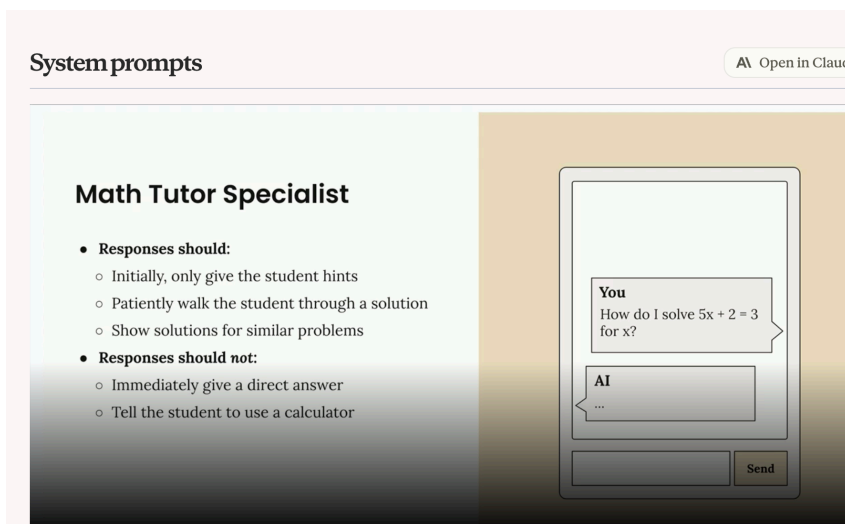
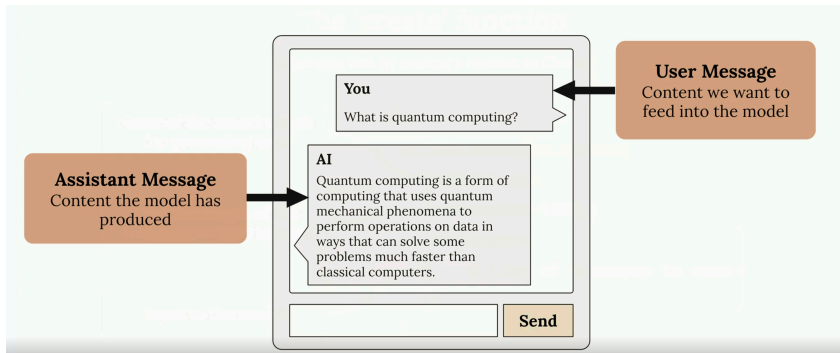
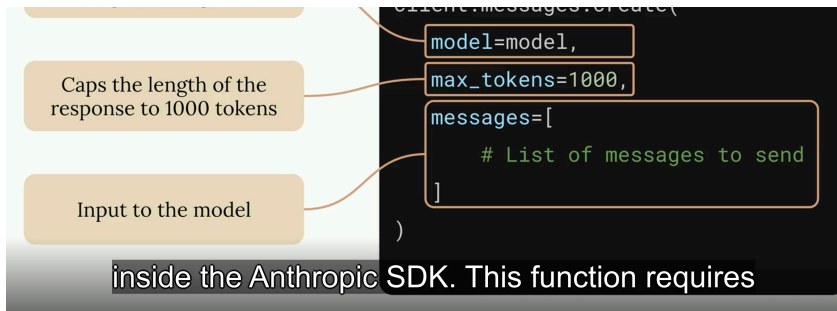


## The 'create' function

Generates text by making a request to Claude

Name of the model to use for generating text

```
client.messages.create(
```



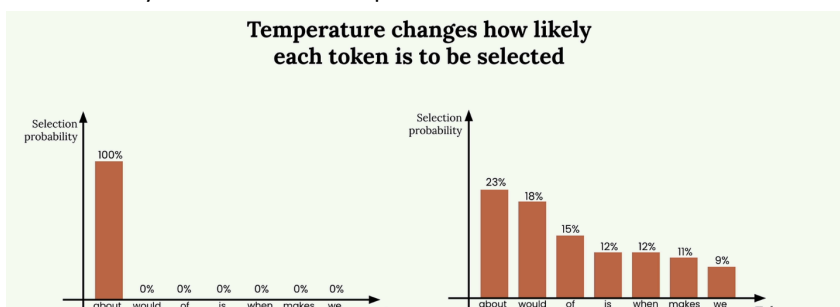
System prompts :- Customizes tone and style of Claude response

## Temperature :-

Claude's text generation process. When you send Claude a prompt like, it goes through three key steps:

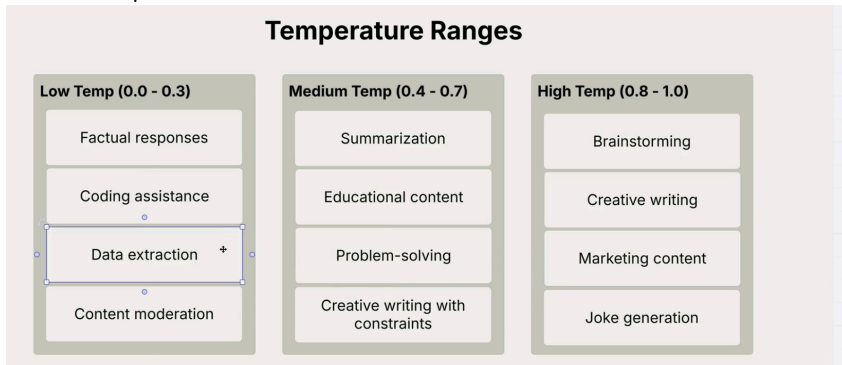
- Tokenization - Breaking your input into smaller chunks
- Prediction - Calculating probabilities for possible next words
- Sampling - Choosing a token based on those probabilities

Temperature is a decimal value between 0 and 1 that directly influences these selection probabilities. It's like adjusting the "creativity dial" on Claude's responses.

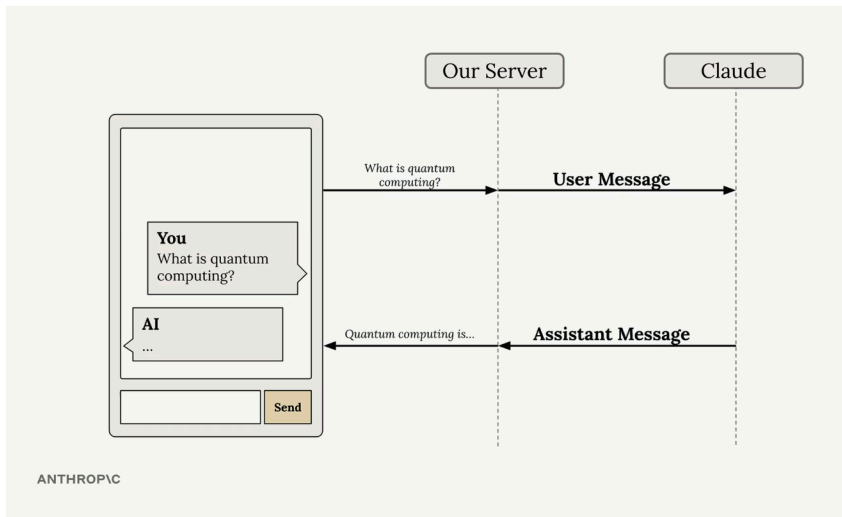




At low temperatures (near 0), Claude becomes very deterministic - it almost always picks the highest probability token. At high temperatures (near 1), Claude distributes probability more evenly across options, leading to more varied and creative outputs.

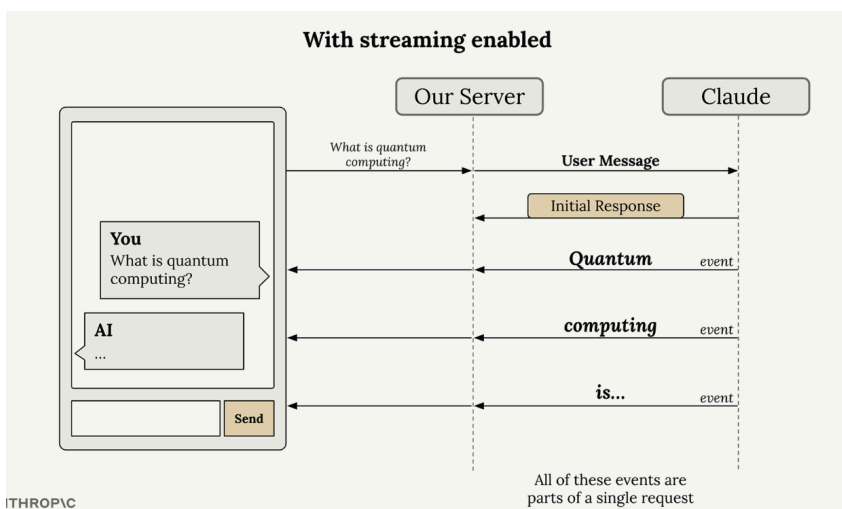


## Response Streaming



When building chat applications with Claude, there's a significant user experience challenge: responses can take 10-30 seconds to generate, leaving users staring at a loading spinner. The solution is response streaming, which lets users see text appear chunk by chunk as Claude generates it, creating a much more responsive feel.

With streaming enabled, Claude immediately sends back an initial response indicating it has received your request and is starting to generate text. Then you receive a series of events, each containing a small piece of the overall response.



Common event responses that arrive

|                                         | MessageStart      | A new message is being sent                                   |
|-----------------------------------------|-------------------|---------------------------------------------------------------|
|                                         | ContentBlockStart | Start of a new block, which contains text, tool use, or other |
| These contain the actual generated text | ContentBlockDelta | Chunk related to the latest content block that was started    |
|                                         | ContentBlockDelta | Chunk related to the latest content block that was started    |
|                                         | ContentBlockDelta | Chunk related to the latest content block that was started    |
|                                         | ContentBlockStop  | The current content block has been completed                  |
|                                         | MessageDelta      | The current message is complete                               |
|                                         | MessageStop       | End of information about the current message                  |

Consider building a web app that generates AWS EventBridge rules. Users enter a description, click generate, and expect to see clean JSON they can immediately copy and use.

[illegible]